Database Design Project: Online Retail Platform

Introduction

This project focuses on designing a database for an online retail platform similar to DigiKala, a leading e-commerce platform. Create and document the database schema required to manage users, products, categories, orders, payments, reviews, inventory, carts, and discounts. Apply database design principles, including the creation of an Entity-Relationship Diagram (ERD), table definitions, and the implementation of an Entity-Attribute-Value (EAV) model for handling variable product attributes, to address real-world e-commerce functionality.

Functional Requirements

1. Users

Users can:

- Register as customers.
- Create and manage multiple addresses for delivery.
- Place orders for products.
- Leave reviews and ratings for products.
- Add products to a shopping cart.
- Save and apply discount codes to orders.
- View order history and track order status.
- Customers can upgrade to a premium account for benefits

2. Products

Products can:

- Belong to multiple categories.
- Have metadata such as title, description, price, upload date, and stock quantity…
- Include variable attributes (e.g., color, size, storage capacity) managed via an EAV model.
- Be rated and reviewed by users.

- Be added to carts or ordered by customers.

3. Categories

Categories can:

- Contain multiple products, and each product can belong to multiple categories.

4. Orders

Orders can:

- Include multiple products
- Be associated with a user, shipping address, and optional discount
- Have a status (e.g., Pending, Shipped, Delivered, Cancelled)
- Be linked to a payment record

5. Payments

Payments can:

- Be associated with an order.
- Include details such as amount, payment date, status, and payment method.
- Support multiple payment methods (e.g., credit card, online banking).

6. Reviews

Reviews can:

- Be submitted by users for products.
- Include a rating and comment.

7. Cart

Carts can:

- Be associated with a user.
- Contain multiple products with quantities.

8. Inventory

Inventory can:

- Track the stock quantity for each product.

9. Discounts(bonus)

Discounts can:

- Be applied to orders using a discount code.
- Have metadata such as code, percentage, start date, and end date.
- Be tracked for usage per user or order.

Database Design Requirements

Students are required to:

1. **Design an ERD**:
    - Model all entities and their relationships.
2. **Define Tables**:
    - Create SQL table definitions for all entities, ensuring proper data types
    - Include tables for the EAV model to manage variable product attributes (e.g., Entity, Attribute, Value).
3. **Implement EAV Model**:
    - Ensure the EAV structure allows adding new attributes without modifying the schema.
4. **Deliverables**:
    - ERD diagram
    - SQL file with table definitions and sample data(Insert Data).
    - Documentation explaining the design

Good Luck:))